Code: MCS101

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

DATA STRUCTURES & ALGORITHM ANALYSIS

(Computer Science and Engineering) 15 APR 2013

Time : 3 hours   Max. Marks: 60

Answer all the questions
All Questions carry equal marks
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1(a) Explain queue applications [6M]

(b) Calculate time complexity for the following algorithm
Algorithm unique elements (A[0..n-1])// checks whether all the elements in a given array are distinct

```
for i < - 0 to n – 2 do
  for j <- i + 1 to n – 1 do
      if A [ i ] = A [ j ]
            return false    return true
```
[6M]

(OR)

(c) Explain we insert a node at the beginning and a + required position in a singly linked list. [6M]

(d) Write post fix notation for the following infix notation. Show all stack operations.   a + b * c + (d * e + f) * g [6M]

2(a) Explain quick sort algorithm with an example. Write its complexity. [12M]

(OR)

(b) Explain insertion sort algorithm with an example. [6M]

(c) Write short notes on extendible hashing. [6M]

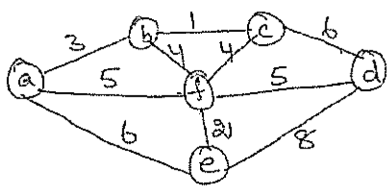
3(a) Define BST. Implement the operation on BST with neat examples. [12M]

(OR)

(b) Explain DFS algorithm [6M]

(c) Explain Biconnected components. [6M]

4(a) Write an algorithm to construct a minimum spanning tree. Using Kruskal's algorithm write minimum spanning tree for the following graph. [12M]



(OR)

(b) Explain strassen's matrix multiplication [6M]

(c) Define greedy technique. Explain how we solve a knapsack problem using greedy method with an example. [6M]

5(a) Explain optimal binary search trees [12M]

(OR)

(b) Explain graph coloring problem [6M]

(c) Explain sum of subsets problem with a neat example. [6M]

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

17 APR 20[illegible]

DATABASE MANAGEMENT SYSTEMS
(Computer Science and Engineering)

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
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1(a) Who are the different types of data base end users? Discuss the main activities of each. [6M]
(b) What are the responsibilities of a DBA [6M]

(OR)

(c) Describe the three-schema architecture [6M]
(d) What are the different types of interfaces provided by a DBMS [6M]

2(a) Explain specialization and generalization [6M]
(b) Explain modeling of union types [6M]

(OR)

(c) Discuss the types of single-level ordered indexes [12M]

3(a) Explain the different categories of constraints on databases [12M]

(OR)

(b) What are the basic data types available in SQL [6M]
(c) Explain the function of EXISTS and UNIQUE in SQL [6M]

4(a) Define 2NF and 3NF. Give suitable examples for each [6M]
(b) Explain Tuple relational calculus [6M]

(OR)

(c) What is multi valued dependency? What type to constraint does it specify. [6M]
(d) Explain lossy and lossless joint decompositions with examples.[6M]

5(a) State the properties of Transactions [6M]
(b) Describe the four levels of Isolation in SQL [6M]

(OR)

(c) What is a dead lock? Explain dead lock prevention methods [6M]
(d) What is the two-phase locking protocol? How does it guarantee serializability? [6M]

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Code: MCS103

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

COMPUTER ORGANIZATION 20 APR 2013
(Computer Science and Engineering)

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

Q.No.	Question	Marks Allotted
	Unit-I	
1(a)	Discuss about Logic Microoperations	6
(b)	Explain about hardware implementation of shift operations.	6
	OR	
(c)	The following transfer statements specify a memory. Explain the memory operation in each case. (a) $R_2 \leftarrow M[AR]$ (b) $M[AR] \leftarrow R_3$ (c) $R_5 \leftarrow M[R_5]$	6
(d)	Explain about address sequencing.	6
	Unit-II	
2 (a)	Explain about instruction cycle	12
	OR	
(b)	Discuss about basic computer design with a neat flowchart.	12
	Unit-III	
3 (a)	Explain about RISC.	8
(b)	A computer has 32bit instructions and 12bit address. If there are 250 two-address instructions. How may one address instructions be able to be formulated?	4
	OR	
(c)	Explain about hardware implementation of divide algorithm with a neat example.	12

Q.No.	Question	Marks Allotted
	Unit-IV	
4 (a)	A computer employs RAM chips of 256 x 8 and ROM chips of 1024 x 8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units, each with four registers. A memory-mapped I/O configuration is used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers. a. How many RAM and ROM chips are needed? b. Draw a memory-address map for the system. c. Give the address range in hexadecimal for RAM, ROM, and interface.	12
	OR	
(b)	A virtual memory system has an address space of 8K words, a memory space of 4K words, and page and block sizes of IK words. The following page reference changes occur during a given time interval. (Only page changes are listed. If the same page is referenced again, it is not listed twice.) 4 2 0 1 2 6 1 4 0 1 0 2 3 5 7 Determine the four pages that are resident in main memory after each page reference change if the replacement algorithm used is (a) FIFO; (b) LRU.	12
	Unit-V	
5 (a)	A DMA controller transfers 16-bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 2400 characters per second. The CPU is fetching and executing instructions at an average rate of 1 million instructions per second. By how much will the CPU be slowed down because of the DMA transfer?	8
(b)	Explain about character oriented and bit oriented protocols.	4
	OR	
(c)	Explain about Input-Output Processor.	6
(d)	Discuss about Daisy-chain priority interrupt.	6

Code: MCS104

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

COMPUTER NETWORKS

(Computer Science and Engineering)

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Explain the TCP/IP reference model [6M]
(b) Explain about ARQANET? [6M]

(OR)

(c) Explain the different guided transmission medium? [12M]

2(a) Explain the CRC with an example? [6M]
(b) Explain any one of the CSMA/CD and CSMA/CA protocols [6M]

(OR)

(c) Explain the frame format of 802.3 protocol? [6M]
(d) Draw the Manchester & differential Manchester encoding for the given message – 1011100111100? [6M]

3(a) Explain about the link state routing protocol? [6M]
(b) Explain the open-loop & closed-loop solutions for controlling congestion? [6M]

(OR)

(c) Explain the need of transparent and non-transparent fragmentation? [6M]
(d) Explain & discuss the IP packet header? [6M]

4(a) Explain the process of symmetric connection release & the 2-army problem? [6M]
(b) Explain the TCP header? [6M]

(OR)

(c) Define silly window syndrome and the solutions to overcome the syndrome. [8M]
(d) Explain about the UDP? [4M]

5(a) Discuss about the SMTP? [6M]
(b) Explain about the pretty good privacy protocol? [6M]

(OR)

(c) Explain in detail about the WWW? [12M]

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Code: MCS1051

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

SOFTWARE ENGIEERING

(Computer Science and Engineering)

Time : 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Explain software myths [6M]
(b) What frame work activities are used during PSP and TSP? Discuss the objectives of each one. [6M]

(OR)

(c) Explain evolutionary process models. [12M]

2(a) Explain Analysis and Design modeling principles in detail. [12M]

(OR)

(b) Explain requirements engineering tasks [12M]

3(a) How we create a behavioral model explain with a neat example. [12M]

(OR)

(b) Explain design concepts in detail [12M]

4(a) Explain data design at component level [6M]
(b) Explain architectural patterns [6M]

(OR)

(c) Explain golden rules for user interface design [12M]

5(a) Write short notes on
(i) Validation testing (ii) Block testing [12M]

(OR)

(b) Explain metrics for the design model. [12M]

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LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

AUTOMATA AND COMPILER DESIGN

(Computer Science and Engineering)

Time : 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Construct DFA accepting all strings over {a, b} ending in ab [6M]
(b) What are the phases in compiler and define them [6M]

(OR)

(c) State and prove Arden's theorem [6M]
(d) Write a short note pass and phases of translation in compiler [6M]

2(a) Define ambiguity grammar and explain with example. [6M]
(b) For the grammar $S \rightarrow aB/bA$ $A \rightarrow aS/bAA$ $A \rightarrow a$, $B \rightarrow bS/aBB/b$ for the string aaabbabbba, find the leftmost and rightmost derivation trees. [6M]

(OR)

(c) What is top-down parsing write the sequence of parse trees for the input id + id * id is a top-down parse according to the grammar $E \rightarrow TE^{|}$ $E^{|} \rightarrow + TE^{|}/E$ $T \rightarrow FT^{|}$ $T^{|} \rightarrow * FT^{|}/E$ $F \rightarrow (E)$ / id [12M]

3(a) Write LR parsing algorithm [6M]
(b) Write a brief note on handle pruning [6M]

(OR)

(c) Construct SLR parsing table for the following grammar
$E \rightarrow E + T/T$ $T \rightarrow t * F/F$ $F \rightarrow (E)$ /id [12M]

4(a) What is syntax directed translation [4M]
(b) Convert into post fix notation for the given infix notation with production $E \rightarrow E_{|} + T$ { print f ('+') ; } $E \rightarrow T$ [8M]

(OR)

(c) Explain the classes S – attribute definition and L-attribute definition of syntax directed definition (SDD) [12M]

5(a) Explain about DAG representation of basic blocks [12M]

(OR)

(b) Write a short note on pointer assignment and procedure calls [6M]
(c) Draw a DAG for the basic blocks x = a[i] a[j] = y z = a[i] [6M]

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Code: MCS1063

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

ARTIFICIAL INTELLIGENCE

(Computer Science and Engineering)

Time : 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) What is artificial intelligence? and its applications [4M]
(b) Discuss problem characteristics [8M]

(OR)

(c) How can you solve the problem? How can you built the system and discuss the water jug problem [12M]

2(a) What is graph? Explain the advantages disadvantages of BFS over DFS. [6M]
(b) Explain A* algorithm? [6M]

(OR)

(c) Explain constraint satisfaction problem and its algorithm? [12M]

3(a) Explain the different approaches to knowledge representation? [12M]

(OR)

(b) Explain the clarification algorithm with suitable examples. [12M]

4(a) Explain about Neural networks [6M]
(b) Discuss about different learning approaches? [6M]

(OR)

(c) Describe the learning decision trees? [6M]
(d) Describe the instance based learning? [6M]

5(a) Explain about syntactic analysis? [6M]
(b) Discuss the grammar induction [6M]

(OR)

(c) Explain the probability language models? [6M]
(d) Explain the probabilistic language processing? [6M]

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Code: MEC101

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

ADVANCED DIGITAL SIGNAL PROCESSING 15 APR 2013
(Systems and Signal Processing (ECE))

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) State and prove the frequency shifting property of DFT. [6M]
(b) Compute the DFT of the following sequence using DIF-FFT algorithm
$x(n) = \{1,1,1,1,1,1,1,1\}$ [6M]

(OR)

(c) Write a short note on filter design for IIR interpolators and decimators. [8M]
(d) What is the need for multistage filter implementation? [4M]

2(a) What is the need for spectral estimation? What are the limitations of Non-parametric methods in spectral estimations. [8M]
(b) Advantages & Disadvantages of Non-parametric power spectrum estimation. [4M]

(OR)

(c) If the sample sequence of a random process has N=2400 samples.
(i) Determine the frequency resolution of the Bartlett, Welch (50% overlap) & Black man-Tukey methods for a quality factor Q = 6.
(ii) Determine the record length for the Bartlett, Welch (50% overlap) & Blackman-Tukey methods. [12M]

3(a) Explain the Music algorithm [6M]
(b) Derive the relation between Auto correlation and spectral density [6M]

(OR)

(c) Consider the ARMA process generated by the difference equation
$x(n) = 1.6\, x(n-1) - 0.63\, x(n-2) + w(n) + 0.9\, w(n-1)$.
(i) Determine the system function of the whitening filter & its poles and zeros.
(ii) Determine the power density spectrum of x(n) [12M]

4(a) What are the properties of the linear prediction filters. [6M]
(b) Explain about Levinson Durbin Algorithm. [6M]

(OR)

(c) Determine the reflection coefficient K_3 in terms of the auto correlations $\{\gamma_{xx}(m)\}$ from the Schur algorithm and compare your result with the expression K_3 obtained from the Levinson Durbin Algorithm. [12M]

5(a) Describe the propagation of input quantization noise to digital filter. [6M]
(b) Explain the quantization of fixed-point numbers. [6M]

(OR)

(c) Explain the two kinds of limit cycle behavior in DSP? [12M]

Code: MEC102

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

17 APR 2013

TRANSFORM TECHNIQUES
(Systems and Signal Processing (ECE))

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Define Fourier transform pair and define conditions for its existence and hence explain the frequency analysis of signals with the help of Fourier transform. [6M]
(b) Discuss about Walsh and Haar transforms [6M]

(OR)

(c) Explain about Hilbert transform properties and applications? [6M]
(d) Explain the relation between Fourier and z-transform techniques. [6M]

2(a) What are the drawbacks of FT? How these can be overcome using STFT? What are properties of STFT [6M]
(b) What is wavelet? Discuss clearly the features of wavelet transform and hence distinguish between continuous and discrete wavelet transforms [6M]

(OR)

(c) Define MRA scaling function and explain the concept of basis for wavelet subspace using Haar wavelet [6M]
(d) Discuss about
(i) Mexican Hat Meyer (ii) Doubechies wavelet transforms [6M]

3(a) Explain about DWT filter banks for Daubenchies wavelet function. [6M]
(b) Explain about two channel filter bank and condition for perfect reconstruction. [6M]

(OR)

(c) Explain the implementation of decomposition filters from which bring out the concept of PRQMF. [6M]
(d) Define orthogonal wavelets and bring out their relation to filter banks.[6M]

4(a) Explain about lifting scheme of wavelet generation. [6M]
(b) Explain about 2-dimensional wavelet. [6M]

(OR)

(c) Describe the filtering relationship for Bi-orthogonal filters. [6M]
(d) Construct wavelet basis functions for a two level balanced tree starting with a Haar scaling function and wavelet. [6M]

5(a) Explain about signal compression using DCT and 2-D DWT? [6M]
(b) Explain about various signal denoising techniques. [6M]

(OR)

(c) Explain about subband coding of speech and music. [6M]
(d) Explain about fractional signal analysis. [6M]

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

VLSI TECHNOLOGY AND DESIGN 20 APR 2013
(Systems and Signal Processing (ECE))

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Derive the relation between drain to source current I_{ds} versus drain to source voltage V_{ds} in saturation and non-saturation region. [6M]
(b) Determine the pullup pull down ratio of an n MOS inverter when driven through one (or) more pass transistors. [6M]

(OR)

(c) Explain the CMOS inverter DC characteristics. [6M]
(d) Explain in detail about MOS transistor with neat sketches. [6M]

2(a) Explain about layout design tools [6M]
(b) Explain the scalable design rules applied to COMS inverter [6M]

(OR)

(c) Explain the power consumption analysis of low power gate with neat diagram. [6M]
(d) Discuss the switch logic implementation of a four way multiplexer. [6M]

3(a) Discuss about power optimization [6M]
(b) Design a layout diagram for two input COMS NAND gate. [6M]

(OR)

(c) How fault simulation is done? Give examples for fault models. [6M]
(d) Explain with an example to calculate the propagation delay in a network and suggest methods to reduce the delay? [6M]

4(a) Explain the clocking strategies? [6M]
(b) Explain about the DRAM cell with read, write operations. [6M]

(OR)

(c) What are the problems presented by power distribution? How they are solved? [6M]
(d) Explain about forming arrays of memory cells. [6M]

5(a) Explain the various routing methods in floor planning? [6M]
(b) Explain about the features of system on chip and Embedded CPU's? [6M]

(OR)

(c) Explain about high level synthesis? [6M]
(d) Explain the issues for the Architecture of low power VLSI [6M]

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Code: MEC104

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

MICROCONTROLLERS FOR EMBEDDED SYSTEM DESIGN

(Systems and Signal Processing (ECE))

Time : 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks

******

1(a) Define the term system and Embedded system. Explain the steps needed to design an Embedded system? [6M]
(b) With block diagram, explain components of embedded system hardware? [6M]

(OR)

(c) Explain about different classification of embedded systems? [6M]
(d) What is an embedded software, explain? [6M]

2(a) What is a memory mapped I/O and explain about memory controller? [6M]
(b) Explain about (i) Microprocessors (ii) Microcontroller. [6M]

(OR)

(c) Draw the block diagram of 8051 microcontroller and explain it? [10M]
(d) Write about memory arbitration? [2M]

3(a) Write the important differences between RISC and CISC processors? [6M]
(b) Explain about different modes on ARM processor? [6M]

(OR)

(c) Write a short nodes on (i) Timber blocks (ii) Digital blocks [6M]
(d) Explain the importance of PSOC? [6M]

4(a) What is a device driver and how it is useful for programmable timing devices? [6M]
(b) Define interrupt and explain how interrupt can be handeled? [6M]

(OR)

(c) Differentiate context and context switching? [6M]
(d) What is a deadline and interrupt latency explain with an example? [6M]

5(a) What are the difference popular protocols used for wireless communication explain? [6M]
(b) Explain various types of bus architecture serial communication protocols? [6M]

(OR)

(c) Explain how Bluetooth devices can be used to setup personal area networks? [6M]
(d) Explain about IDMA? [6M]

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Code: MEC1052

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

IMAGE AND VIDEO PROCESSING

(Systems and Signal Processing (ECE))

Time : 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Explain the fundamental steps in digital image processing. [6M]
(b) Describe the basic concepts in sampling & quantization. [6M]

(OR)

(c) Explain in detail 2D-Discrete Fourier transform. [6M]
(d) Explain the relationship between pixels. [6M]

2(a) Explain the sharpening spatial filtering. [6M]
(b) Describe the smoothing frequency domain filters. [6M]

(OR)

(c) Explain the point & line detection techniques. [6M]
(d) Explain thresholding. [6M]

3(a) Explain coding Redundancy. [6M]
(b) Describe Huffman coding. [6M]

(OR)

(c) Explain the transform coding system (i) encoder (ii) Decoder. [6M]
(d) Write about wavelet coding. [6M]

4(a) Explain the basic steps of video processing. [6M]
(b) Compare Geometric image formation with photometric image formation. [6M]

(OR)

(c) Explain three dimensional time-varying image formation models. [6M]
(d) In detail explain filtering operations. [6M]

5(a) Describe block matching algorithm. [6M]
(b) Explain with necessary diagrams about Region based motion estimation. [6M]

(OR)

(c) Write about waveform based coding estimation. [6M]
(d) Explain the application of motion estimation in video coding. [6M]

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LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

RADAR SIGNAL PROCESSING

(Systems and Signal Processing (ECE))

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks

******

1(a)	Draw and explain about general radar block diagram and explain radar range performance.	[7M]
(b)	Explain radar detection with noise jamming.	[5M]
	(OR)	
(c)	Explain matched filter for non-white noise	[5M]
(d)	Explain in detail about the characteristics and its derivation of correlation function and correlation detection.	[7M]
2(a)	Draw and explain in detail of Likelihood-ratio receiver.	[6M]
(b)	Write the difference between envelop and Logarithmic detectors.	[6M]
	(OR)	
(c)	Draw and explain of inverse probability receiver.	[6M]
(d)	Draw and explain of CFAR receiver and write the uses in radar.	[6M]
3(a)	Explain Ambiguity function with example in the design of radar waveforms and explain about the ambiguity principles and properties.	[6M]
	(OR)	
(b)	Explain the design requirements of optimum waveforms for detection in clutter.	[12M]
4(a)	Draw and explain of Linear FM pulse compression in detail	[7M]
(b)	Write the stretch techniques	[5M]
	(OR)	
(c)	Explain in detail of SAW pulse compression.	[5M]
(d)	Explain generation and decoding of FM waveforms in detail with neat diagram.	[7M]
5(a)	Draw and explain of phase coded-CW Radar and write the application of CW radar.	[7M]
(b)	Explain about Binary phase code, Barker codes.	[5M]
	(OR)	
(c)	Write short notes on (i) Maximal length sequences (MLS) (ii) Non-linear FM pulse compression (iii) Linear period modulation (LPM)	[12M]

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Code: MEE101

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Supplementary Examinations, October 2012

MACHINE MODELING AND ANALYSIS 15 APR 2013
(Power Electronics and Drives(EEE))

Time: 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) For a two pole DC machine develop circuit model. Derive the armature voltage and field expression for a DC machine. Express the above equation in matrix form. [12M]

(OR)

(b) Prove that for a torque to produced by the interaction of two quadrature eminent in the generalized rotating electric machine. [8M]

(c) What observations are made from the impedance matrix of this machine. [4M]

2(a) Develop the mathematical model for a separately excited DC motor explain how they can be expressed in state variable form. Obtain the transfer function for no-load and on-load operations. [12M]

(OR)

(b) Explain the mathematical model of DC series and shunt motors [12M]

3(a) Discuss in detail about the three phase to two phase transformation. (a,b,c to α, β, o) [12M]

(OR)

(b) Explain the importance of linear transformation for a three phase induction motor.
Discuss about (a) phase transformation (b) A transformation to a common reference frame with a special reference to three phase induction motor. [12M]

4(a) Determine the two a ruls model for induction motor. Obtain voltage and current equations in rotor reference frame. [12M]

(OR)

(b) Deduce expression for voltage and torque of symmetrical two phase induction motor in station reference frame. [12M]

5(a) Obtain transient torque characteristics of synchronous motor with approximated mathematical expression. [12M]

(OR)

(b) Obtain the circuit model of a three phase synchronous motor discuss in detail about two-axis representation of a three phase synchronous motor. [12M]

Code: MEE102

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

POWER CONVERTERS
(Power Electronics and Drives (EEE)) 17 APR 2013

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
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1(a) The three-phase bidirectional delta-connected controller has a resistive load of R = 10 Ω. The line – to – line voltage is V_s = 208 V (rms), 60 HZ, and the delay angle is $\alpha = 2\pi/3$ Determine (a) the rms output phase voltage V_0 ; (b) the expressions for instantaneous currents i_a, i_{ab}, i_{ca} ; (c) the rms output phase current I_{ab} and rms line current I_a ; (d) the input PF ; and (e) the rms current of a thyristor I_R.

(OR)

(b) Explain the 1-ϕ voltage controller with inductive loads and derive the expression for rms output voltage and current and rms and average thyristor currents.

2(a) Write short notes on
(i) Extriction angle control (ii) Single-phase sinusoidal PWM

(OR)

(b) Derive the expression for three-phase half-wave converters with neat circuit diagrams and waveforms.

3(a) Discuss about the power circuit principle of operation

(OR)

(b) Write short notes on the following
(i) Single phase stage boost power factor corrected rectifier
(ii) Three phase boost PFC converter.

4(a) Write short notes on the following.
(b) Space rector modulation (ii) Current source inverters

(OR)

(b) Briefly explain
(i) Modified sinusoidal PWM (ii) Variable dc-link inverter.

5(a) Explain the principle of operation of cascaded multilevel inverters and write it features.

(OR)

(b) What is multilevel inverter and discuss the concept of multilevel inverter and list out the advantages and disadvantages of diode-clamped multilevel inverter.

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Code: MEE103

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Supplementary Examinations, ~~October 2012~~ 20 APR 2013

CONTROL OF MOTOR DRIVES-I
(Power Electronics and Drives (EEE))

Time: 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Briefly discuss the steady state analysis of a three phase fully controlled converter fed series DC motor drives. [12M]

(OR)

(b) Explain the reduction of harmonic using multiple-pulse control. [12M]

2(a) Briefly explain the steady state analysis of chopper controlled DC motor drives. [12M]

(OR)

(b) Write short notes on the following
(i) Rating of the devices (ii) Model of the chopper. [12M]

3(a) The single phase half bridge inverter has the DC input of 48 V. The load resistance is 4.8 Ω, Determine (i) RMS value of output voltage (ii) RMS value of fundamental component (iii) Total harmonic distortion. [12M]

(OR)

(b) Write short notes on
(i) Harmonic factor of η^{th} harmonic (ii) Total harmonic distortion
(iii) Distortion factor (iv) Lowest order harmonic [12M]

4(a) Briefly explain the speed and flux control is current source inverter. [12M]

(OR)

(b) Explain efficiency optimization control by flux program. [12M]

5(a) Write short notes on
(i) Static Kramer drive (ii) Static scherbius drive [12M]

(OR)

(b) Explain the following
(i) Rotor resistance control (ii) Slip power recovery schemes. [12M]

Code: MEE104

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
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M.Tech. I Semester Supplementary Examinations, October 2012

SPECIAL MACHINES

(Power Electronics and Drives (EEE))

Time: 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Explain the factor effecting the torque to weight ratio of SRM [6M]
(b) Explain the closed loop control of steps motor [6M]

(OR)

(c) Derive an expression to find the developed torque in a stepper motor[6M]
(d) Explain the design procedure of stator and rotor pole arcs. [6M]

2(a) Explain the properties of a permanent magnet [6M]
(b) Why electric commutation can be called as spark in commutation. [6M]

(OR)

(c) Give the constructional details of rotor of a BLDC motor [6M]
(d) What is the effect of magnetic saturation is a permanent magnet DC motor [6M]

3(a) Explain the operation of single sided LIM [6M]
(b) What is the influence of end effect on the thrust developed in a LIM. [6M]

(OR)

(c) What are the operational difference of short stator and long stator LIM configuration. [6M]
(d) What are the realistic consideration of LIM [6M]

4(a) Give the constructional details of PM linear synchronous motor [6M]
(b) Develop the equivalent circuit of air core linear synchronous motor[6M]

(OR)

(c) Explain how thrust in developed in a permanent magnet linear synchronous motor [6M]
(d) Give the difference in operation between iron core LSM and PMLSM[6M]

5(a) Classify the servo motor on the basic of operation [6M]
(b) Derive the transfer function of a field controlled DC servo motor. [6M]

(OR)

(c) Give the constructional details of DC servo motor [6M]
(d) Explain the application of AC servo motor with reasoning. [6M]

Code: MEE1051

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Supplementary Examinations, October 2012

MODERN CONTROL THEORY

(Power Electronics and Drives (EEE))

Time: 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks

******

1(a) List the advantages of Non-linear analysis [6M]

(b) Derive the describing function of a ideal relay with dead zone nonlinearity. [6M]

(OR)

(c) Explain how the existence of limit cycles and their nature can be predicted using Describing function analysis. [6M]

(d) What are the limitations of describing function approach. [6M]

2(a) Using delta method develop the phase plane trajectory for the system 'X'+1x1+x+x^3=0; with x(0) = -1; x(0) = 0.5 [12M]

(OR)

(b) What are the advantages of Isodine method [6M]

(c) Explain the significance of closed trajectories formed in the phase plane. [6M]

3(a) What is the significance of positive definite and positive semi definite function. [6M]

(b) Determine a suitable liapunor function for investigating the stability of the following system [6M]

$X^0 + X^0 + X^2 - 1 = 0$

(OR)

(c) Explain Kraskorskii's method for constructing the liapunor's function[6M]

(d) Check the sign differentiate of the function:

(i) $x_1^2 + x_2^2 + x_1.x_2$ (ii) $(x_1+x_2)^2$ [6M]

4(a) Explain the characteristics of an observable system. [6M]

(b) Give the procedure to design a static feedback controller through pole placement. [6M]

(OR)

(c) Consider the system: $\dot{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \mu$ and y = [2 -1] x

Design a reduced under state observer that makes the estimation error to decay at least as fast as e^{-10t} [12M]

5(a) Explain the minimum energy problem [6M]

(b) Explain low tracking problem can be solved using calculator variations. [6M]

(OR)

(c) Explain the state regulator problem. [6M]

(d) Explain the procedure to design state regulation through liapunor equation. [6M]

Code: MEE1061

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

DIGITAL CONTROLLERS FOR POWER ELECTRONIC SYSTEMS
(Power Electronics and Drives (EEE))

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Briefly explain the PIC reset Actions and PIC memory organizations [12M]

(OR)

(b) Explain the PIC oscillator connections, addressing modes and I/O ports. [12M]

2(a) Explain the functional block diagram of DSP 2407 controller [12M]

(OR)

(b) Explain the mapping external devices to the C2xx core and instruction sets. [12M]

3(a) Explain the interrupt Hierarchy and interrupt control registers. [12M]

(OR)

(b) Write short notes on the following
(i) Initializing and servicing interrupts in software.
(ii) General purpose I/O overview [12M]

4(a) Briefly explain the capture UNITS and quadrature enclosed pulse circuits. [12M]

(OR)

(b) Generate a code for fixed duty cycle PWM signal using DSP 2407 controller with comments. Explain various sub-components present in EV module of DSP 2407 controller. [12M]

5(a) Write HDL code to implement EX-OR gate and NAND gate using LEDs in SPARTAN 3 E. [12M]

(OR)

(b) Write short notes on the following.
(i) Configurable logic Blocks (CLB)
(ii) Input / Output Block (IOB) [12M]

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Code: MIT101

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Supplementary Examinations, October 2012

REQUIREMENTS ENGG. & ESTIMATION
(Software Engineering (IT))

Time: 3 hours   Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Define software requirements. What are the levels of requirements. [8M]
(b) Explain about the process maturity [4M]

(OR)

(c) Explain the benefits of high quality requirements. [6M]
(d) Explain about the practical process improvement. [6M]

2(a) Explain requirement analysis and its activities. [6M]
(b) Explain any two requirements analysis model. [6M]

(OR)

(c) What are the requirements elicitation techniques. Explain the techniques with merits and demerits. [12M]

3(a) Explain about the testing of requirements with examples. [6M]
(b) Explain about software requirements management [6M]

(OR)

(c) Explain about the software inspection process. [6M]
(d) Explain about the requirements engineering for critical system. [6M]

4(a) What are the two views of sizing? Explain about the function point analysis. [12M]

(OR)

(b) Explain the following:
(i) Mark II FPA   (ii) LOC estimation [6+6M]

5(a) Write a short notes on COCOMO II [6M]
(b) Explain the desirable features in software estimation tools. [6M]

(OR)

(c) Write a short note on Putnam estimation model. [6M]
(d) Explain the features of requirements rational pro. [6M]

Code: MIT102

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

SOFTWARE ARCHITECTURES 17 APR 2013
(Software Engineering (IT))

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) What is software architecture? Explain the status of software architecture. [12M]

(OR)

(b) What makes a good software architecture [6M]
(c) Why software architecture is important [6M]

2(a) Write about the architectural structures for shared information systems [12M]

(OR)

(b) Discuss the tools used for integration in software development environments. [12M]

3(a) Explain the statistical analysis techniques useful in analysis design with a quantified design space [12M]

(OR)

(b) Explain about architectural for malison? What are its applications, merits & demerits [12M]

4(a) Write about the requirements of architectural description language [12M]

(OR)

(b) What are traditional programming languages. Explain about adding implicit invocation to traditional programming languages. [12M]

5(a) Describe the WRIGHT model of architectural description. [12M]

(OR)

(b) Write about exploiting steps in architectural design with applications, merits & demerits. [12M]

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Code: MIT103

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M.Tech. I Semester Supplementary Examinations, ~~October 2012~~ 20 APR 2013

SOFTWARE PROCESS MANAGEMENT
(Software Engineering (IT))

Time: 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Distinguish between software process & Software project. Discuss in detail the initial process. Repeatable process & Managed process. [12M]

(OR)

(b) What is meant by software process maturity? Explain process maturity levels with neat diagram. [12M]

2(a) What are the actions required to establish a project management system.
(b) Write about the software product nomenclature [6+6M]

(OR)

(c) Write about the project planning principles
(d) Mention various critical elements of software quality management system. [6+6M]

3(a) Explain briefly the following
(i) Quality indicators (ii) Issue tracking (iii) Reviews & inspections. [12M]

(OR)

(b) Discuss about quality assurance & configuration management.
(c) Differentiate between standards & guidelines. [6+6M]

4(a) Write about the software quality management [12M]

(OR)

(b) What is software measurement? How can we classify them. Discuss various software size measures. [12M]

5(a) Discuss the costs & benefits of defect prevention
(b) Write about the software process changes for defect prevention. [6+6M]

(OR)

(c) Write about the organizational plans to automate technology transaction & productivity. [12M]

Code: MIT104

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

WEB SEARCHING AND MINING

(Software Engineering (IT))

Time : 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Explain about the indexing and keyword search [6M]
(b) Explain about document Representation [6M]

(OR)

(c) Explain about the Relevance Ranking with examples, merits and demerits [12M]

2(a) Explain about the Hierarchical agglomerative clustering with examples. [12M]

(OR)

(b) Explain about similarity based criterion function with advantages and disadvantages. [12M]

3(a) What is classification. Explain about the general settings and evaluation techniques. [12M]

(OR)

(b) Write about the feature selection with examples, merits & demerits [12M]

4(a) Write about the cross-industry standard process for data mining. [12M]

(OR)

(b) Write about the user identification in data preprocessing [12M]

5(a) Write about the calculation of average time per page and duration for individual pages. [12M]

(OR)

(b) Write about the BRICH clustering algorithm [12M]

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Code: MIT1052

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

MOBILE COMPUTING

(Software Engineering (IT))

Time : 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

******

1(a) Explain about mobile computing architecture with neat diagram [6M]
(b) Describe the mobile computing limitations [6M]

(OR)

(c) Explain briefly about cellular telephone system [6M]
(d) What are the functions of mobile computing [6M]

2(a) Explain about network switching subsystem in GSM briefly [6M]
(b) Describe the four handover scenarios in GSM [6M]

(OR)

(c) Explain about TDMA with neat diagram and describe the advantages of TDMA [6M]
(d) Explain about the hidden /exposed terminal problems [6M]

3(a) Explain about mobile IP packet delivery with neat diagram [6M]
(b) Explain about agent advertisement in mobile IP agent discovery [6M]

(OR)

(c) Explain about the traditional TCP and what are its advantages and disadvantages [6M]
(d) Explain briefly about fast retransmission and fast recovery [6M]

4(a) Explain about IEEE 802.11 system architecture briefly [6M]
(b) Explain about physical layer of IEEE 802.11 briefly [6M]

(OR)

(c) Explain briefly about DSDV routing protocol and describe its advantages and disadvantages [6M]
(d) Explain the properties of mobile adhoc networks [6M]

5(a) Explain briefly about WAP architecture and describe all the layers of WAP [6M]
(b) What are the characteristics and features of WAP protocol [6M]

(OR)

(c) Explain about blue tooth system architecture briefly [6M]
(d) What are the user scenarios of blue tooth [6M]

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Code: MIT1063

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

DIGITAL IMAGE PROCESSING

(Software Engineering (IT))

Time : 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) What is digital image processing? What the fundamental steps in digital image processing? What are the components of image processing system? [12M]

(OR)

(b) Explain image quantization & image sampling in detail [12M]

2(a) Explain in detail the Histogram modeling with examples. [12M]

(OR)

(b) Explain about weiner filtering and explain its digital implementation [12M]

3(a) Explain the following
(i) Median filters (ii) Homomorphic filters [12M]

(OR)

(b) Explain different color transformation [6M]
(c) Explain about JPEG compression in detail [6M]

4(a) Explain the following morphological operations with suitable example.
(i) Pruning (ii) Opening & Closing (iii) Erosion (iv) Dilation [12M]

(OR)

(b) What is image segmentation? Explain it using threshold region based segmentation technique. [12M]

5(a) Write a short note on
(i) Patterns (ii) Pattern classes (iii) Adaptive learning systems [12M]

(OR)

(b) Explain how image recognition is done based on decision – theoretic methods [12M]

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Code: MME101

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

15 APR 2013

ADVANCED MATHEMATICS
(Thermal Engineering (ME))

Time : 3 hours — Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) (i) Explain ill and well-conditioned linear system of equations with example also [5M]
(ii) Solve the following system of equations using LU decomposition method.

$x + 5y + z = 14$
$2x + y + 3z = 13$
$3x + y + 4z = 17$ [7M]

(OR)

(b) Explain diagonally dominant principle. Also solve the following system by Gauss-Seidel method

$28x + 4y - z = 32$
$x + 3y + 10z = 24$
$2x + 17y + 4z = 35$ [12M]

2(a) Find the dominant eigen value and the corresponding eigen vector of

$$A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Find also the least latent root and hence the third eigen vector.[12M]

(OR)

(b) From the following table, find the value of x for which f(x) is a maximum. Also find the maximum value of f(x) from the table of values given below.

x	: 60	75	90	105	120
f(x)	: 28.2	38.2	43.2	40.9	37.7

[12M]

3(a) Using Romberg's method, evaluate
$I = \int_0^1 \frac{1}{1+x} dx$, correct to three decimal places. Hence evaluate log e^2 [12M]

(OR)

(b) Solve $\frac{dy}{dx}$ = 1 – y ; y (0) = 0 using modified Euler's method and tabulate the solutions at x = 0.1, 0.2 and 0.3. Compare your results with the exact solutions. [12M]

4(a) Explain initial and boundary value problems also solve the equation $y^{11} = x + y$ with the boundary conditions y (0) = y(1) = 0 [12M]

(OR)

(b) Apply Runge-Kutta method to find approximate value of y for x=0.2, is steps of 0.1, if $\frac{dy}{dx} = x + y^2$, given that y = 1 where x = 0 [12M]

5(a) Solve the equation $\nabla^2 u = -10 (x^2 + y^2 + 10)$ over the square with sides x = 0, y = 0, y = 3, x = 3 with u = 0 on the boundary and mesh length = 1 [12M]

(OR)

(b) Using crank-Nicholson method, solve $u_{xx} = 16 u_t$, $0 < x < 1$, $t > 0$, given that u (x, 0) = 0, u (0, t) = 0, u (1, t) = 100 t.
Compute u for one step in t directing taking h = ¼. [12M]

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M.Tech. I Semester Regular/Supplementary Examinations

ADVANCED THERMODYNAMICS
(Thermal Engineering (ME))

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks
******

1(a) Establish the equivalence of Kelvin-Planck and clausius statements. [6M]

(b) A perfect reversed heat engine is used for making ice at -5^0C from water available at 25^0C. The temperature of freezing mixture is -10^0C. Calculate the quantity of ice formed per KWh.
For ice specific heat = 2.1 kJ/kg K ; latent heat = 335 kJ/Kg. [6M]

(OR)

(c) Show that for Vander waal's eqn for gas

$$(S_2 - S_1)_T = R\, l_n \frac{v_2 - b}{v_1 - b}$$ [6M]

(d) Set up the expression for changes in internal energy for a gas which obeys the Vander waal's equation $(p + a/v^2)(v - b) = RT$ [6M]

2(a) Derive an expression for availability in non flow process? [6M]

(b) Two kg of air at 500 kPa, 80^0C expands adiabatically in a closed system until its volume is doubled and its temperature becomes equal of that of the surroundings which is at 100 kPa, 5^0C for this process. Determine (i) The maximum work (ii) The change in availability? [6M]

(OR)

(c) Water is to be heated at constant pressure from 25^0C to 80^0C. If the heat source is at constant temperature of 500^0C and the ambient temperature is 20^0C. What would be the gain in availability of water and effectiveness of the heating process? For water C_P= 4.18 kJ/kgK [6M]

(d) A quantity of air initially at 1 bar, 300K undergoes two types of interactions (i) It is brought to a final temperature of 500 K adiabatically by paddle wheel work transfer (ii) The same temperature rise in brought about by het transfer from a thermal reservoir at 600 K Take T_0 = 300 K ; P_0 = 1 atm. Determine the irreversibility in kJ/Kg in each case and comment on the result?[6M]

3(a) Determine the adiabatic flame temperature when liquid octane at 25^0C is burned with 300% theoretical air at 25^0C in a steady flow process? [6M]

(b) Explain first law for Reactive systems? [6M]

(OR)

(c) The following data pertains to test run made to determine the calorific value of a sample of coal having hydrogen content of 5% mass of coal burnt 0.85 gm. Mass of fuse wire burnt and its calorific value = 0.028 gm; 6700 kJ/Kg. mass of water in calorimeter = 1800 gm. Initial and Final temperature of water is 16.5^{0}C and 20.25^{0}C. water equivalent of calorimeter is 350 gm ; cooling correction is 0.08^{0}C make calculations for the higher and lower calorific values of the coal sample? [6M]

(d) Explain about calorific value of the fuels? [6M]

4(a) Explain Mach number and compressible flow regimes [6M]

(b) An airoplane moving at a supersonic speed is not beard by stationary observer until the plane has passed him. Comment on the validity of this statement. Measuring the Mach angle is an accurate means of measuring supersonic velocities for sharp nosed objects such as projectiles. Estimate the velocity and mach number of a projectile when Mach cone of much angle π/6 radians is observed from the projectile at an altitude where temperature is 275^{0}K. If the Mach angle is estimated with in ± 4%, Determine the velocity range of the fighter. Take R = 287 J/Kg K and γ = 1.4 [6M]

(OR)

(c) Prove that the sonic velocity for an adiabatic process $a=\sqrt{\gamma RT}$ [6M]

(d) Gas turbine gases leave the nozzle with a stagnation pressure of 6 bar and stagnation temperature of 1200 K. If the static pressure is 1 bar determine the static temperature and velocity of jet. Consider the flow is reversible adiabatic? [6M]

5(a) A turbojet aircraft flies with a vel of 300ms^{-1} at an altitude where the air is at 0.35 bar and – 40^{0}C. The compressor has press ratio of 10 and temperature of gases at turbine inlet is 1100^{0}C. Air enters the compressor at rate of 50 Kg/sec. Estimate (i) The temperature and pressure of the gases at the turbine exit (ii) The velocity of gases at nozzle exit (iii) The propulsive efficiency of cycle. [12M]

(OR)

(b) In a single heater regenerative cycle the steam enters the turbine at 30 bar, 400^{0}C and the exhaust pressure is 0.10 bar. The feed water heater is direct contact type which operates at 5 bar. Find (i) The efficiency and the steam rate of the cycle and (ii) The increase in mean temperature of heat addition efficiency and steam rate. Neglect pump work? [12M]

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Code: MME103

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

ADVANCED HEAT & MASS TRANSFER 20 APR 2013

(Thermal Engineering (MECH))

Time : 3 hours   Max. Marks: 60

Answer all the questions
All Questions carry equal marks

******

1(a) A Refrigerated container in the form of a cube with 2m sides 5mm thick aluminum walls is insulated with a 10 cm layer of cork (K = 0.043 W/mk). During steady operation the temperatures on the inner and outer surfaces of the container are measured to be -5^{0}C and 20^{0}C respectively. Determine the cooling load on the refrigerator. Given 'K' (Thermal conductivity) for aluminum is 204 W/m^{0}K. [6M]

(b) What are the different modes of Heat transfer and explain them? [6M]

(OR)

(c) Derive general heat conduction equation in Cartesian coordinates? [6M]

(d) A current of 200A is passed through a stainless-steel wire (K = 19 Wm^{0}K) 3mm in diameter. The resistivity of the steel may be taken as 70 μ Ω.cm and the length of wire is 1m. The wire is submerged in liquid at 110^{0}C and experiences a convection heat-transfer coefficient of 4 KW/m^2 ^{0}K calculate the centre temperature of the wire? [6M]

2(a) Prove that the temperature of a body at any time τ during Newtonian heating (or) cooling is given by the relation $\frac{t - t_a}{t_i - t_a} = \exp\left(-B_i F_o\right)$ where B_i and F_o are Biot and Fourier modulus respectively. $t_a \rightarrow$ ambient temperature and $t_i \rightarrow$ initial temperature of body [6M]

(b) An egg with mean diameter of 4cm and initially at 25^{0}C is placed in a boiling water pan for 4 minutes and found to be boiled to the consumers taste. For how long should a similar egg for same consumer be boiled when taken from a refrigerator at 5^{0}C? use lumped parameter theory and pressure the following properties for egg.
K = 12W/m^{0}K ; h = 125 W/m^2 ^{0}K ; C = 2 kJ/Kg ^{0}K ρ = 1250 Kg/m^3 [6M]

(OR)

(c) A long aluminum cylinder 5 cm in a diameter and initially at 200^{0}C is suddenly exposed to be a convection environment at 70^{0}C and h = 525 W/m^2 ^{0}C Calculate the temperature at radius of 1.25 cm and the heat lost per unit length 1 minute after the cylinder is exposed to environment. [6M]

(d) An average human body modeled as a 30 cm diameter. 170 cm long cylinder has 72% water by mass, so that its properties may be taken as those of water at room temperature. ρ = 1000 Kg/m^3 C = 4180 J/Kg ^{0}K and K = 0.608 W/m^{0}K. A person is found dead at 5 AM in a room temperature of which is 20^{0}C. The temperature of a body is measured to be 25^{0}C when found, and the heat transfer coefficient is estimated to be 8 W/m^{0}K. Assume the body temperature of a living man is 37^{0}C, Estimate the time of death of the above person? [6M]

3(a) Explain in detail the mechanism of forced convection, show by dimensional analysis that data for forced convection may be correlated by an equation of the form $N_u = \phi$ (R_e, P_r). Where Nusselt number $N_u = \frac{hl}{K}$ Reynolds number $R_e = \frac{Vl\rho}{\mu}$ and Prandtl number $P_r = \frac{\mu\, cp}{K}$ [6M]

(b) Air at 27°C and 1 atm flows over a flat plate at a speed of 2 m/sec calculate the boundary layer thickness at distance of 20 and 40 cm from the leading edge of the plate. Calculate the mass flow which enters the boundary layer between x = 20 cm and x = 40 cm. The viscosity of air at 27°C is 1.85×10^{-5} Kg/m.s Assume unit depth in the Z-direction? [6M]

(OR)

(c) A steam pipe 50 mm diameter and 2.5 m long has been placed horizontally and exposed to still air at 25°C. If the pipe wall temperature is 295°C, Determine the rate of heat loss, at the mean temperature of 160°C, the thermo-physical properties of air are $K = 3.64 \times 10^{-2}$ W/m °K $\nu = 30.09 \times 10^{-6}$ m²/sec ; $P_r = 0.682$ and $\beta = 2.31 \times 10^{-3}$ K⁻¹ [6M]

(d) A copper bus bar of round cross-section with 15 mm diameter is cooled with a cross flow of dry air. The air is at 20° C and it flows past the bus bar with the velocity of 1.5 ms-1 Make calculations for the coefficient of heat transfer from the surface of bus bar to the cooling air. What would be the permissible current intensity for the bus bar if its surface temperature is not to exceed 75°C? Resistivity of copper is 0.0175×10^{-6} ohm-m³/min. For a single cylinder placed in cross flow, the following empirical correlations have been suggested.

$N_u = 0.44\, R_e^{0.5}$ $\quad 10 < R_e < 10^3$

$N_u = 0.22\, R_e^{0.6}$ $\quad 10^3 < R_e < 2 \times 10^5$ [6M]

4(a) What is condensation? Explain film wise condensation and drop wise condensation? [6M]

(b) A vertical square plate, 30 x 30 cm is exposed to steam at atmospheric pressure the plate temperature is 98°C. Calculate the heat transfer and the mass of steam condensed per hour? [6M]

(OR)

(c) Two concentric cylinders having diameters of 10 and 20 cm have a length of 20cm. Calculate the shape factor between the open ends of the cylinder? [6M]

(d) A pipe carrying steam having an outside diameter of 20 cm runs in a large room and is exposed to air at temperature of 30°C. The pipe surface temperature is 200°C. Find the heat loss per metre length of the pipe by convection and radiation taking the emissivity of the pipe surface as 0.8? [6M]

5(a) What is the Mass transfer and explain the classification of Mass transfer process? [6M]

(b) Air at 100°C flow over a stream lined naphthalene body. Naphthalene sublimes into air and its vapour pressure at 100°C is 20mm of Hg. The heat transfer coefficient for this system was previously found to be 17.45 W/m² °K. The mass diffusivity of naphthalene vapour in air at 100°C is 2.98×10^{-2} m²/h. The concentration of naphthalene in bulk air stream is negligible small. Calculate the mass transfer coefficient and the mass flux for the system. For air at 100°C, take $\rho = 0.946$ Kg/m³, $C_p = 1.005$ kJ/KgK K = 0.032 W/m°K and $\nu = 23.13 \times 10^{-6}$ m²/sec. [6M]

(OR)

(c) The air pressure in a tyre tube of surface area 0.5 m² and wall thickness of 0.01m is approximated to drop from 2 bar to 1.99 bar in a period of 5-days. The solubility of air in rubber is 0.07 m³ of air /m³ of rubber at 1 bar estimate the diffusivity of air in rubber at the operating temperature of 300 K if the volume of air in tube is 0.025 m³. [6M]

(d) Gaseous hydrogen is stored at a elevated pressure in a rectangular steel container of 10 mm wall thickness. The molar concentration of hydrogen in steel at the outer surface is 2 Kg mol/m³, while the concentration of hydrogen in steel at the outer surface is 0.5 kg mol/m³. The binary diffusion coefficient for hydrogen in steel is 0.26×10^{-12} m²/sec. what is the mass flux of hydrogen through the steel? [6M]

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Code: MME104

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

22 APR 2013

IC ENGINE COMBUSTION & POLLUTION

(Thermal Engineering (MECH))

Time: 3 hours

Max. Marks: 60

Answer all questions
All Questions carry equal marks

Q.No			Marks
1	a)	Explain the combustion phenomenon in C.I. engines with the help of a neat sketch.	(6)
	b)	Explain the classification of automobiles with respect to different types of engines.	(6)
		OR	
	c)	Explain the working of a Rotary engine with the help of a neat diagram.	(6)
	d)	Explain the working of a flywheel.	(6)
2	a)	What are the requirements of an ignition system for an S.I. Engine?	(6)
	b)	What is detonation? Explain	(6)
		OR	
	c)	Explain the factors that influence the flame speed.	(6)
	d)	What are the characteristics of an ideal combustion chamber?	(6)
3	a)	Explain the various phases of C.I engine combustion.	(6)
	b)	Explain the factors on which delay period depends.	(6)
		OR	
	c)	What is cold starting ? Explain	(6)
	d)	Explain diesel engine injected spray combustion process.	(6)
4	a)	Explain the procedure to measure I.P of a multi cylinder engine.	(6)
	b)	Explain the working of a brake rope dynamometer.	
		OR	
	c)	Explain trans esterification process	(6)
	d)	Explain the advantages and dis advantages of using alcohols as fuels.	(6)
5	a)	Explain the various sources from which pollutants are emitted from C.I. engines.	(6)
	b)	What are the technological considerations to be taken care for a vehicle to be called Euro III compliant?	(6)
		OR	
	c)	Explain the effect of engine load on diesel engine smoke.	(6)
	d)	What are the effects of emission on human health?	(6)

Code: **MME1051**

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

GAS DYNAMICS

(Thermal Engineering (ME))

Time : 3 hours | Max. Marks: 60

Answer all the questions
All Questions carry equal marks

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1(a) Explain limitations on air as a perfect gas? [6M]

(b) A fighter aircreaft attains its maximum speed of 2160 kmph at an altitude of 12 km. The take-off speed at sealevel is 270 kmph. If the flight speed increases linearly with altitude, compute the variation of stagnation temperature with altitude for a climb up to the max speed? [6M]

(OR)

(c) Derive expressions for specific quantities of internal energy and volume in adiabatic flow process? [6M]

(d) Explain second law of thermodynamics for compressible flow? [6M]

2(a) A convergent – divergent diffuser is to be used at Mach 3. The diffuser has to be use a variable throat area so as to swallow the starting shock what % increase in throat area will be necessary[6M]

(b) Calculate the dynamic pressure of the flow if V_∞ = 175 ms^{-1}, P_∞ = 1 atm and T_∞ = 298 K. what will be the percentage error if the flow is treated as incompressible? [6M]

(OR)

(c) A storage chamber compressor is maintained at 1.8 atm absolute and 20^0C. The surrounding ambient pressure is 1 atm. Calculate (i) The velocity with which air flow will take place from the chamber to the outside through a unit area hole (ii) Mass flow rate per unit area. Assume air as perfect gas? [6M]

(d) A Delavel nozzle has to be designed for an exit Mach number of 1.5 with exit diameter of 200 mm. Find the ratio of throat area / exit area necessary. The reservoir conditions are given as P_0 = 1 atm ; T_0 = 20^0C. Find also the maximum mass flow rate through the nozzle what will be the exit pressure and temperature? [6M]

3(a) Explain Applications of shock tubes? [6M]
(b) What is meant by weak shock and strong shock? [6M]

(OR)

(c) A normal shock wave occurs in air flowing at a Mach number of 1.5. The static pressure and temperature of the air upstream of a shock wave are 1 bar and 300 K. Determine the Mach number, pressure and temperature downstream of the shock wave. Also estimate the shock strength? [6M]
(d) Derive the basic Momentum equation for control volume having normal shock wave. [6M]

4(a) Consider a 2-Dimensional duct carrying a perfect gas with uniform conditions P_1 = 1 atm and M_1 = 2. Design a 10^0 turning elbow to achieve a uniform downstream state 2 for each of the following cases?
(i) $M_2 > M$; $S_2 = S_1$ (ii) $M_2 < M_1$; $S_2 = S_1$
(iii) $M_2 < M_1$; $S_2 > S_1$ (iv) $M_2 = M_1$; $S_2 = S_1$
In each case find numerical values for M,P and duct area (Compared to that of state 1) [12M]

(OR)

(b) Explain about Detached shocks? [6M]
(c) Explain about supersonic compression? [6M]

5(a) Atmospheric air at pressure 1.0135 x 10^5 N/m^2 and temperature 300K is drawn through friction-less bell month entrance into a 3m long tube having a 0.05 m diameter. The average friction coefficient $\bar{f}$ = 0.05 for the tube. The system is perfectly insulated. Find the maximum mass flow rate and the range of back pressures that will produce this flow? [6M]
(b) Air at standard sea level conditions enters the tube at Mach 0.68 and reaches a value of mach 0.25 at exit of the diffuser (station B). The entrance area is 1 m^2. Assuming no dissipative losses in the diffuser. Show that the area at station 'B' is 2.16 m^2 will be the area be larger (or) smaller is losses are present? [6M]

(OR)

(c) Explain the concept of flow in constant area duct with friction? [6M]
(d) Explain effect of wall friction on fluid properties? [6M]

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Code: MME1061

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)

L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech.(I Semester) Examinations, October, 2012

SOLAR ENERGY

Time: 3 hours

Max. Marks: 60

Answer all questions

All Questions carry equal marks

Q.No			Marks
1	a)	What is scattering of solar rays? Explain	(6)
	b)	The latitude of Srinagar is 34^0. Find day length in hours when sun light is available on 1st July.	
		OR	
2	a)	Explain the phenomena of extraterrestrial solar radiations.	
	b)	Explain the phenomena of solar radiation on tiled surfaces.	
3	a)	Explain solar energy storage and its applications.	
	b)	What is Transmittance Absorptance product? Explain.	
		OR	
4		Explain the various solar heating techniques.	
5	a)	Explain the advantages and disadvantages of concentrating collectors over flat-plate collectors.	(6)
	b)	What is selective surface? Explain its importance in the performance of solar heaters.	(6)
		OR	
6		Explain Rabl's theorem for Maximum concentration. Derive an expression for Concentration ratio	(12)
7	a)	What are the different methods of solar energy storage?	(4)
	b)	Explain the principle of collection of solar energy used in non-convection solar pond.	(8)
		OR	
8		Explain the working of a forced circulation drier with the help of a neat sketch. Which parameters control the drying rate?	(12)
9	a)	Explain the performance characteristics of a solar cell.	(6)
	b)	Explain the working principle of a solar cell.	(6)
		OR	
10		Derive the expression for calculating the Present Worth Factor(PWF) to determine the economic viability of the solar system.	(12)